28/03/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

PST-2

Time : 180 Mins.

Answers

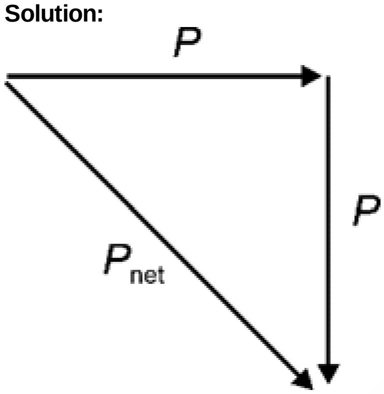
1.	(3)	37.	(1)	73.	(4)	109.	(2)	145.	(1)
2.	(4)	38.	(2)	74.	(1)	110.	(2)	146.	(2)
3.	(2)	39.	(2)	75.	(4)	111.	(3)	147.	(2)
4.	(3)	40.	(4)	76.	(1)	112.	(1)	148.	(3)
5.	(4)	41.	(4)	77.	(1)	113.	(2)	149.	(2)
6.	(2)	42.	(2)	78.	(3)	114.	(4)	150.	(4)
7.	(2)	43.	(2)	79.	(4)	115.	(1)	151.	(2)
8.	(4)	44.	(2)	80.	(1)	116.	(4)	152.	(1)
9.	(4)	45.	(3)	81.	(2)	117.	(2)	153.	(3)
10.	(2)	46.	(2)	82.	(2)	118.	(4)	154.	(2)
11.	(2)	47.	(4)	83.	(2)	119.	(3)	155.	(1)
12.	(4)	48.	(1)	84.	(4)	120.	(3)	156.	(4)
13.	(3)	49.	(4)	85.	(1)	121.	(2)	157.	(1)
14.	(4)	50.	(1)	86.	(4)	122.	(1)	158.	(1)
15.	(3)	51.	(1)	87.	(1)	123.	(2)	159.	(1)
16.	(4)	52.	(1)	88.	(3)	124.	(4)	160.	(3)
17.	(3)	53.	(3)	89.	(1)	125.	(3)	161.	(4)
18.	(4)	54.	(3)	90.	(3)	126.	(2)	162.	(3)
19.	(2)	55.	(2)	91.	(3)	127.	(1)	163.	(3)
20.	(4)	56.	(4)	92.	(3)	128.	(1)	164.	(3)
21.	(1)	57.	(1)	93.	(3)	129.	(3)	165.	(4)
22.	(2)	58.	(3)	94.	(4)	130.	(4)	166.	(4)
23.	(2)	59.	(4)	95.	(2)	131.	(1)	167.	(3)
24.	(1)	60.	(3)	96.	(2)	132.	(3)	168.	(3)
25.	(4)	61.	(2)	97.	(3)	133.	(3)	169.	(4)
26.	(4)	62.	(3)	98.	(3)	134.	(1)	170.	(1)
27.	(2)	63.	(1)	99.	(3)	135.	(2)	171.	(2)
28.	(2)	64.	(4)	100.	(1)	136.	(1)	172.	(3)
29.	(4)	65.	(2)	101.	(2)	137.	(4)	173.	(2)
30.	(1)	66.	(2)	102.	(3)	138.	(2)	174.	(4)
31.	(4)	67.	(1)	103.	(2)	139.	(4)	175.	(2)
32.	(1)	68.	(4)	104.	(2)	140.	(2)	176.	(3)
33.	(4)	69.	(3)	105.	(4)	141.	(1)	177.	(1)
34.	(1)	70.	(2)	106.	(4)	142.	(2)	178.	(2)
35.	(3)	71.	(4)	107.	(3)	143.	(2)	179.	(1)
36.	(4)	72.	(3)	108.	(3)	144.	(1)	180.	(2)

Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
Current is induced in metallic ring such that it opposes the change of flux through it.

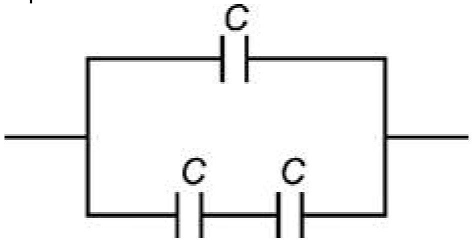
(2) **Answer :** (4)
Solution:
$\phi = 4t^2 - 2t + 1$
$|\varepsilon| = \left|\frac{d\phi}{dt}\right| = 8t - 2$
at $t = 1$ s, $|\varepsilon| = 8 - 2 = 6$ V
$i = \frac{|\varepsilon|}{R} = \frac{6}{5} = 1.2$ A

(3) **Answer :** (2)
Solution:


$P_{net} = \sqrt{P^2 + P^2}$
$= \sqrt{2}P$

(4) **Answer :** (3)
Solution:
$P_{net} = P_1 + P_2$
$20 = \frac{V^2}{R_1} + \frac{V^2}{R_2}$
$20 = \frac{100}{10} + \frac{100}{R_2}$
$\Rightarrow \frac{100}{R_2} = 10$
$\Rightarrow R_2 = 10\ \Omega$

(5) **Answer :** (4)
Solution:
Capacitance of an isolated conducting sphere is $4\pi\varepsilon_0 R$

(6) **Answer :** (2)
Solution:
Equivalent circuit can be redrawn as


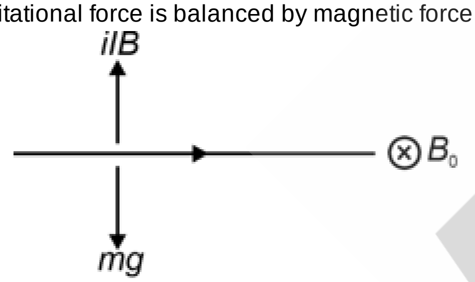
$C_{eq} = \frac{C}{2} + C = \frac{3C}{2}$

(7) **Answer :** (2)
Solution:
Work done W = $U_f - U_i$
3 = $-MB\cos60° - (-MB)$
3 = $-\frac{MB}{2} + MB$
3 = $\frac{MB}{2}$
⇒ MB = 6 unit
Now, Torque τ = $MB\sin60°$
= $6 \times \frac{\sqrt{3}}{2}$
= $3\sqrt{3}$ N m

(8) **Answer :** (4)
Solution:
$T = 2\pi\sqrt{\frac{I}{MB}}$
⇒ $\frac{T^2MB}{4\pi^2} = I$
⇒ $\frac{100\times25\times10^{-4}\times4\times10^{-4}}{4\times10} = I$
= 25×10^{-7} kg m^2

(9) **Answer :** (4)
Solution:
Torque on a loop in magnetic field is given by
$\tau = \overrightarrow{M} \times \overrightarrow{B}$
= $iAB\sin0°$
= Zero

(10) **Answer :** (2)
Solution:
Gravitational force is balanced by magnetic force

⇒ 

⇒ $ilB = mg$
$B = \frac{mg}{il}$
⇒ $B = \frac{50\times10^{-3}\times10}{4\times1}$
= 125×10^{-3} T

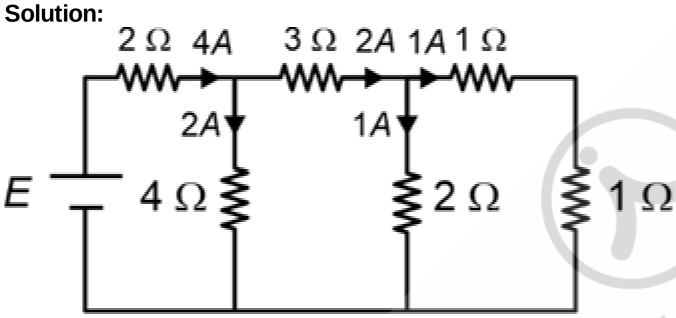
(11) **Answer :** (2)
Solution:
Inductive reactance $X_L = \omega L$
= $2\pi fL$
= $2\pi \times 50 \times \frac{2}{\pi} \times 10^{-3}$
= 200×10^{-3}
= 0.2 Ω

(12) **Answer :** (4)
Solution:
Flux $\phi = \overrightarrow{B} \cdot \overrightarrow{A}$
$= \left(\hat{i} - 2\hat{j} + 4\hat{k}\right) \cdot \left(2\hat{i} - 2\hat{j}\right)$
= 2 + 4
= 6 Wb

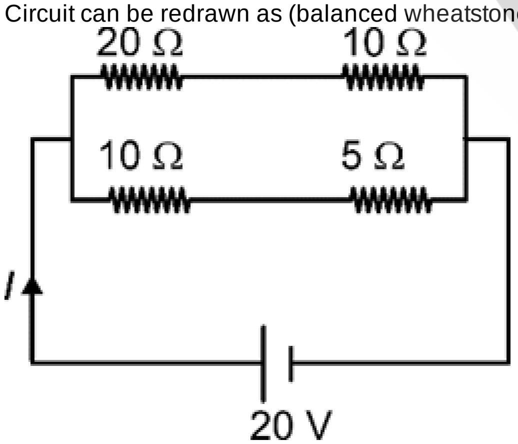
(13) **Answer :** (3)
Solution:
In case of solid sphere,
$V_{\text{centre}} = \frac{3}{2}V_{\text{surface}}$
$= \frac{3}{2} \times 10$
= 15 V

(14) **Answer :** (4)
Solution:
$E = \frac{-\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j}$
$E = -12\hat{i} - (-2y)\hat{j}$
$E_{\text{at }(1,-1)} = \left(-12\hat{i} - 2\hat{j}\right)$ V m^{-1}
$|E| = \sqrt{144+4} = \sqrt{148}\ \frac{\text{V}}{\text{m}}$

(15) **Answer :** (3)
Solution:
Gauss's law is based on the inverse-square dependence on distance inherent in Coulomb's law and is true for any closed surface.

(16) **Answer :** (4)
Solution:


R_{eq} = 4 Ω and current through battery i = 4 A
∴ E = 4 × 4
= 16 V

(17) **Answer :** (3)
Solution:
Circuit can be redrawn as (balanced wheatstone bridge)


$i_{\text{net}} = \frac{20}{15} + \frac{20}{30} = \frac{60}{30} = 2$ A

(18) **Answer :** (4)
Solution:
When B_2 is fused, then potential difference across B_1 decreases and across B_3 increases.

(19) **Answer :** (2)
Solution:
$\vec{F}_m = q(\vec{v} \times \vec{B})$

Magnetic field due to wire at position of charge q is along $\left(-\hat{k}\right)$

$\vec{F}_m \left|\right| \left(\hat{j} \times (-\hat{k})\right)$

$\Rightarrow \vec{F}_m \left|\right| \left(-\hat{i}\right)$

(20) **Answer :** (4)
Solution:
Force on a current carrying closed loop placed in uniform magnetic field is zero

$\therefore \vec{F}_{PQ} + \vec{F}_{QR} + \vec{F}_{RP} = 0$

$0 + \vec{F}_{QR} + \vec{F} = 0$

$\vec{F}_{QR} = -\vec{F}$

(21) **Answer :** (1)
Solution:

$P_S = \frac{V_S^2}{R}$

$R \propto \frac{1}{P_S}$

$\therefore$ Resistance of filament of 20 W bulb is maximum and in series combination $P \propto R$.

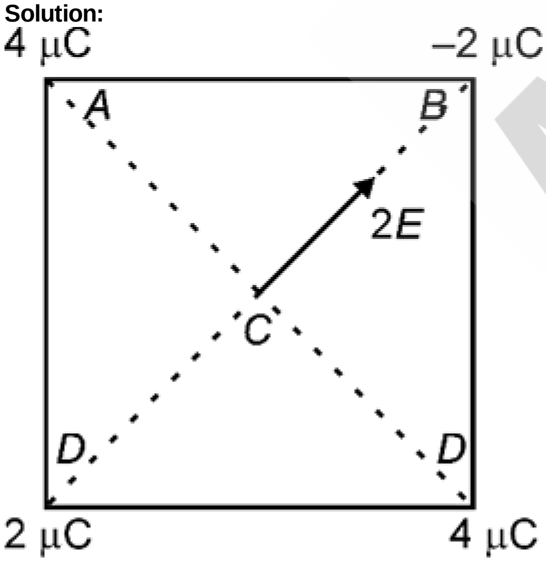
(22) **Answer :** (2)
Solution:
Inductive reactance $X_L = \omega L$
$= 100 \times 0.1$
$= 10\ \Omega$

$i_{rms} = \frac{\varepsilon_{\text{rms}}}{X_L}$

$= \frac{100}{10}$

$= 10$ A

(23) **Answer :** (2)
Solution:



$E_{\text{net}} = 2E$

$= \frac{2kq}{r^2}$

$= \frac{2 \times 9 \times 10^9 \times 2 \times 10^{-6}}{3^2}$

$= 4 \times 10^3$ N

(24) **Answer :** (1)
Solution:
Net electric field inside the cavity will be zero due to shielding.

⇒ Net charge induced on inner surface of cavity will be zero.

(25) Answer : (4)
Solution:
Diamagnetic – Feebly repelled by magnet
Paramagnetic – Feebly attracted by magnet
Ferromagnetic – Strongly attracted by magnet

(26) Answer : (4)
Solution:
Magnetic susceptibility $\chi = \frac{I}{H}$
∴ Unitless and dimensionless

(27) Answer : (2)
Solution:
In transformer,
$\frac{N_S}{N_P} = \frac{V_S}{V_P} = \frac{i_P}{i_S}$
⇒ $\frac{1000}{500} = \frac{24}{i_S}$
⇒ i_S = 12 A

(28) Answer : (2)
Solution:
$|\varepsilon| = \frac{d\phi}{dt}$
⇒ $\int iRdt = \int d\phi$
⇒ $R\int idt = \Delta\phi$
⇒ $5 \times \frac{1}{2} \times 10 \times 0.2 = \Delta\phi$
⇒ $\Delta\phi = 5$ Wb

(29) Answer : (4)
Solution:
$X_L = \omega L$
= 0.04 × 1000
= 40 Ω
Power factor cosθ = $\frac{R}{\mathbb{Z}}$
= $\frac{30}{\sqrt{30^2+40^2}}$
$= \frac{3}{5}$
= 0.6

(30) Answer : (1)
Solution:
Conductors have equipotential surface and electric field lines are always perpendicular to equipotential surface.

(31) Answer : (4)
Solution:
$E = E_1 \sin\omega t + E_2 \cos\omega t$
$E = \sqrt{E_1^2 + E_2^2}\ \sin(\omega t + \varphi)$
$E_{rms} = \sqrt{\frac{E_1^2+E_2^2}{2}}$

(32) Answer : (1)
Solution:
Heat produced by direct current, $H_1 = i^2Rt$
$= 4Rt$
Heat produced by alternating current, $H_2 = i_{rms}^2 Rt$
$= 2Rt$

$\therefore \frac{H_1}{H_2} = \frac{4Rt}{2Rt}$

$= \frac{2}{1}$

(33) Answer : (4)

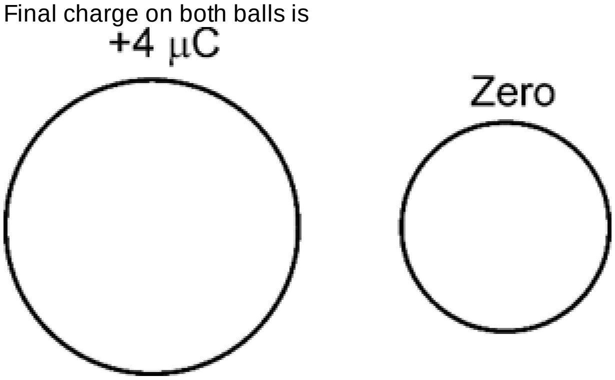
Solution:

When glass rod loses its electrons, then it acquires positive charge.

(34) Answer : (1)

Solution:

Final charge on both balls is



$\therefore F$ = Zero

(35) Answer : (3)

Solution:

$\varepsilon_{eq} = 2E$

$R_{eq} = 3 + r$

$i = \frac{\varepsilon_{eq}}{R_{eq}} = \frac{2E}{3+r}$

Potential difference measured by voltmeter $V = E - ir$

$\frac{E}{4} = E - \frac{2E}{3+r} \cdot r$

$\Rightarrow \frac{2Er}{3+r} = \frac{3E}{4}$

$\Rightarrow 8r = 9 + 3r$

$5r = 9$

$r = \frac{9}{5}\ \Omega$

(36) Answer : (4)

Solution:

Direction of magnetic field due to solenoid is along the axis of solenoid. So magnetic force is zero.
Therefore, the electron will move with same velocity along the axis of solenoid.

(37) Answer : (1)

Solution:

Initially P.D. across both capacitors is same. Now dielectric slab is inserted between plates of capacitor C_1 and thus its capacitance increases.

Therefore, $V = \frac{Q}{C}$, in series $V \propto \frac{1}{C}$

$\therefore V_1 < V_2$

(38) Answer : (2)

Solution:

$\tau_{res} = \tau$

$C\theta = NiAB$

Current sensitivity = $\frac{\theta}{i} = \frac{NAB}{C}$

(39) Answer : (2)

Solution:

Force on current carrying wire $\vec{F} = i(\vec{l} \times \vec{B})$

$\vec{F} = 2(4 \times 10^{-2}\,\hat{i} \times 2(-\hat{k}))$

$= \left(16 \times 10^{-2}\hat{j}\right)$ N

(40) Answer : (4)

Solution:

Since reading of voltmeter is 100 V, that means circuit is in resonance.

So, $\omega = \frac{1}{\sqrt{LC}}$

$2\pi f = \frac{1}{\sqrt{LC}}$

⇒ $4\pi^2 f^2 = \frac{1}{LC}$

⇒ $C = \frac{1}{4\pi^2 f^2 L}$ = $\frac{1}{4\times 10\times 50\times 50\times 1}$

$C = \frac{1}{100\times 10^3}$ = 10 μF

(41) Answer : (4)

Solution:

Magnetic field due to a long straight wire at a distance *r* is given by

$B = \frac{\mu_0 i}{2\pi r}$

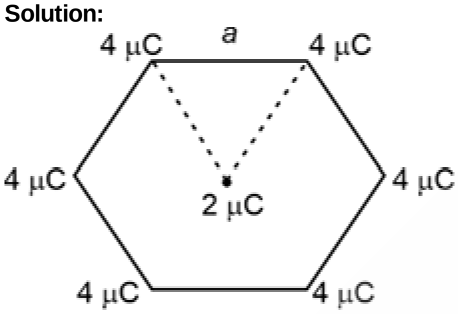
$= \frac{4\pi\times 10^{-7}\times 8}{2\pi\times 2\times 10^{-2}}$

= 8×10^{-5} T

= 80 μT

(42) Answer : (2)

Solution:



$V_i = 0$

$V_f = \frac{6\times kq}{a}$

= $\frac{6\times 9\times 10^9\times 4\times 10^{-6}}{0.9}$

⇒ $V_f = 24 \times 10^4$ V

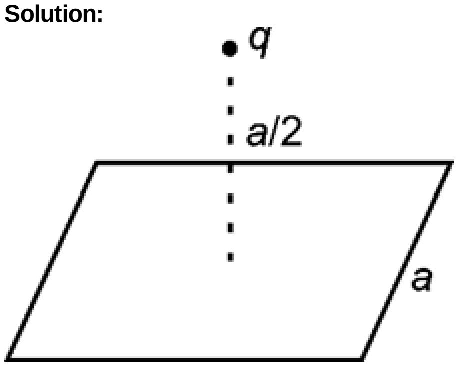
Work done = $q(V_f - V_i)$

= $2 \times 10^{-6}[24 \times 10^4 - 0]$

= 48×10^{-2} = 0.48 J

(43) Answer : (2)

Solution:



Electric flux through square $\phi = \frac{1}{6}\left(\frac{q}{\varepsilon_0}\right)$

= $\frac{8.85\times 10^{-6}}{8.85\times 10^{-12}\times 6}$

$= \frac{10^6}{6}$

$= 1.67 \times 10^5$ V m

(44) Answer : (2)

Solution:

Apply charge conservation

$Q_i = Q_f$

$\Rightarrow 200 \times 10^{-6} \times 100 = 600 \times 10^{-6} \times V_C$

$\Rightarrow V_C = \frac{200}{6}$ V

$= \frac{100}{3}$ V

(45) Answer : (3)

Solution:

If n identical cells are connected in series in which 2 are connected in reverse, then

$\varepsilon_{eq} = (n - 4)E$

$\therefore \varepsilon_{eq} = (20 - 4)E$

$= 16E$

CHEMISTRY

(46) Answer : (2)

Solution:

No. of mole = m × $W_{solvent}$

0.25 = 2XW

W = 0.125 kg = 125

(47) Answer : (4)

Solution:

If $d_{solution}$ = 1 g/ml

Then molarity of 1 molal solution will be less than 1.

Mole fraction is temp independent.

(48) Answer : (1)

Solution:

Gases with more solubility has less value of K_H

He → 144.97 bar^{-1}

N_2 → 76.48 bar^{-1}

CO_2 → 1.67 bar^{-1}

CH_2CHCl → 0.611 bar^{-1}

(49) Answer : (4)

Solution:

$P_S = P_A^0 X_A + P_B^0 X_B$

$= 200 \times \frac{0.42}{0.42 + 0.94} + 415 \times \frac{0.94}{0.42 + 0.94}$

= 200 × 0.31 + 415 × 0.69

= 348.35

$X_{CHCl_3} = \frac{62}{348.35}$

= 0.18

(50) Answer : (1)

Solution:

As ΔT_f increases freezing point of solution decrease.

Hence 0.1 M NaCl having minimum ΔT_f will have maximum freezing point.

(51) Answer : (1)

Solution:

0.1 M NaCl (aq) and 0.1 m KCl will have same boiling point if they are getting 100% ionised.

(52) **Answer :** (1)
Solution:
For isotonic solution
$\pi_1 = \pi_2$
$i_1C_1 = i_2C_2$
For $CaCl_2$ ⇒ iXC = 2.6 × 0.1 = 0.26
For 0.13 M KCl ⇒ iXC = 0.13 × 2 = 0.26

(53) **Answer :** (3)
Solution:
For spontaneous cell
$\Delta G < 0$, $E_{cell} > 0$

(54) **Answer :** (3)
Solution:
$$\mathrm{E_{cell}} = \mathrm{E^{\circ}_{cell}} - \frac{0.0591}{\mathrm{n}}\log\frac{[\mathrm{Zn}^{2+}]}{[\mathrm{H}^+]^2}$$
$$= 0.76 - \frac{0.0591}{n}\log\frac{10^{-2}}{10^{-8}}$$
= 0.76 – 0.1773
= 0.5827 V

(55) **Answer :** (2)
Solution:
Ag is less reactive than Cu.

(56) **Answer :** (4)
Solution:
Given cell is a concentration cell for which
$E^{\circ}_{cell} = 0$
So, $K_{eq} = 1$

(57) **Answer :** (1)
Solution:
Specific conductance k decreases with dilution due to decrease in no. of ion per ml of solution.

(58) **Answer :** (3)
Solution:

Cell	Anode
Leclanche cell	Zn
Mercury cell	Zn-Hg amalgam
Lead storage Battery	Pb
Nickel-Cadmium cell	Cd

(59) **Answer :** (4)
Solution:
$2H_2O \rightarrow 4H^+ + O_2 + 4e^-$
n-factor for O_2 = 4
hence charge required = 4 F

(60) **Answer :** (3)
Solution:
$r = K[A]^2[B]^{-1}$
On making conc. Twice
$R' = K[2A]^2[2B]^{-1}$
$r' = 4 \times \frac{1}{2} K[A]^2[B]^{-1}$
$r' = 2r$

(61) **Answer :** (2)
Solution:
$r = k[X]^a[Y]^b$

$1.2 \times 10^{-3} = k(0.1)^a(0.2)^b$
$3 \times 10^{-4} = k(0.1)^a(0.1)^b$
$2.4 \times 10^{-3} = k(0.2)^a(0.2)^b$
(i) ÷ (ii)
$2^b = 4$
b = 2
(i) ÷ (iii)
$\left(\frac{1}{2}\right)^a = \frac{1}{2}$
a = 1
$r = k[X][Y]^2$

(62) **Answer :** (3)
Solution:
It is a zero order reaction
$t_{100\%} = \frac{a}{k}$
$= \frac{2}{2.5 \times 10^{-2}}$
= 80 s

(63) **Answer :** (1)
Solution:
For first order
r = K[A]
0.001 = K × 0.1
$K = 10^{-2}$
$t_{\frac{1}{2}} = \frac{0.693}{K} = \frac{0.693}{10^{-2}} = 69.3\ s$

(64) **Answer :** (4)
Solution:
$t_{99.9\%} = 10\ t_{\frac{1}{2}}$
= 10 × 10
= 100 min

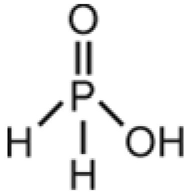
(65) **Answer :** (2)
Solution:
Hydrolysis of ester in acidic medium is a pseudo first order reaction.
Decomposition of NH_3 on Pt(s) surface at high pressure is zero order reaction.

(66) **Answer :** (2)
Solution:
Rate constant increases with increase in temperature for both exothermic and endothermic reaction.

(67) **Answer :** (1)
Solution:

Oxide	Nature
N_2O	Neutral
N_2O_3	Acidic
As_2O_3	Amphoteric
Bi_2O_3	Basic

(68) **Answer :** (4)
Solution:
Basicity of H_3PO_2 is 1



(69) Answer : (3)
Solution:
Order of boiling point of group 16 hydrides is
$H_2S < H_2Se < H_2Te < H_2O$

(70) Answer : (2)
Solution:
Order of negative electron gain enthalpy of group 16 elements is
S > Se > Te > O

(71) Answer : (4)
Solution:
F_2 has less bond dissociation enthalpy due to lone pair repulsion.
Order of bond dissociation enthalpy is
$I_2 < F_2 < Br_2 < Cl_2$

(72) Answer : (3)
Solution:
HClO has less acidic strength than $HClO_4$.

(73) Answer : (4)
Solution:
XeF_6 is sp^3d^3 hybridised with one l.p, of electron on Xe atom hence shape will be distorted octahedral.

(74) Answer : (1)
Solution:

Species	No. of unpaired electron(n)	$\mu = \sqrt{n(n+2)}$
Sc^{3+}	0	0
Ni^{2+}	2	$\sqrt{8}$
Fe^{3+}	5	$\sqrt{35}$
Co^{2+}	3	$\sqrt{15}$

(75) Answer : (4)
Solution:
$[Sc(H_2O)_6]Cl_3$ is d^2sp^3 hybridised with octahedral shape.

(76) Answer : (1)
Solution:
$[Ti(H_2O)_6]^{3+}$ has one unpaired electron that absorb visible light due to d-d transition which results into colour.

(77) Answer : (1)
Solution:
$KMnO_4 + I^{\ominus} \xrightarrow{H^+} Mn^{2+} + I_2$
Oxidation state of Mn is changing from +7 to +2.

(78) Answer : (3)
Solution:
$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$

(79) Answer : (4)
Solution:
$$\underset{\text{Orange}}{K_2Cr_2O_7} \xrightarrow{OH^-} \underset{\text{Yellow}}{K_2CrO_4}$$

(80) Answer : (1)
Solution:
Pr, Nd, Tb and Dy can show +4 oxidation state.

(81) Answer : (2)
Solution:
No is an actinoid.

(82) **Answer :** (2)
Solution:
$[Co(ox)_3]^{3-}$ can exhibit optical isomerism as it does not have any plane of symmetry.

(83) **Answer :** (2)
Solution:
As synergic bonding increases metal carbon bond strength increases for metal carbonyls.

(84) **Answer :** (4)
Solution:
Correct order of field strength of ligands is
$edta^{4-} < NH_3 < en < CN^{\ominus}$

(85) **Answer :** (1)
Solution:
Since H_2O is strong field ligand with Co^{3+}, pairing of d-electrons will take place.
Configuration of Co^{3+} in $[Co(H_2O)_6]Cl_3$ according to Crystal field theory will be $t_{2g}^6\ e_g^0$

(86) **Answer :** (4)
Solution:
Group-I – Dilute HCl
Group-II – H_2S + dilute HCl
Group-III – $NH_4OH + NH_4Cl$
Group-IV – $H_2S + NH_4OH$

(87) **Answer :** (1)
Solution:
Heat evolved on neutralisation of 1 mol NaOH with 1 mol HCl is nearly 57.1 kJ.

(88) **Answer :** (3)
Solution:
For preliminary examination of Cl^- concentrated H_2SO_4 is used.

(89) **Answer :** (1)
Solution:
Lyophilic sol → Egg albamin, starch and gum
Lyophobic sol → Ferric hydroxide, Aluminium hydroxide and arsenic sulphide

(90) **Answer :** (3)
Solution:

Sr. No.	Name of indicator	Colour in acidic medium	Colour in basic medium	Working pH range of indicators
1.	Methyl orange	Orange red	Yellow	3.1 to 4.5
2.	Methyl red	Red	Yellow	4.2 to 6.2
3.	Phenol red	Yellow	Red	6.2 to 8.2
4.	Phenolphthalein (HPh)	Colourless	Pink	8.2 to 10.2

BOTANY

(91) **Answer :** (3)
Solution:
Pollen grain has an outer wall layer called exine, made up of sporopollenin.

(92) **Answer :** (3)
Solution:
Yellow pod colour, terminal position of flower, constricted pod shape and wrinkled seed shape are recessive traits, which are expressed only in homozygous condition.

(93) **Answer :** (3)
Solution:

Klinefelter's syndrome is caused due to the presence of an additional copy of X-chromosome, resulting in a karyotype of 47, XXY. Polygenic inheritance is exemplified by human skin colour.

(94) Answer : (4)
Solution:
A shoot apex and few leaf primordia enclosed in a hollow foliar structure is called the coleoptile.

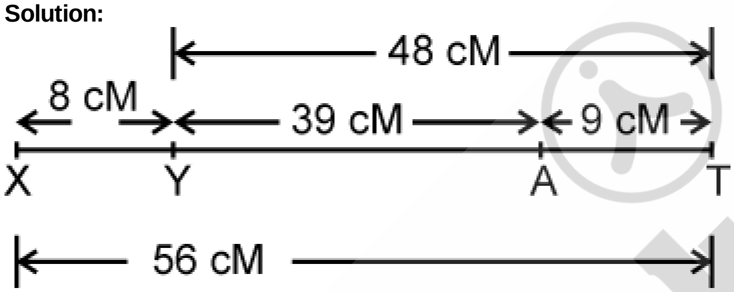
(95) Answer : (2)
Solution:
Viola (common pansy) and *Oxalis* produce two types of flowers – cleistogamous flowers and chasmogamous flowers. Cleistogamous flowers produce assured seed set.

(96) Answer : (2)
Solution:
Geitonogamy is the transfer of pollen grains from the anther to the stigma of another flower of same plant. Although it is functionally cross-pollination involving a pollinating agent but genetically it is similar to autogamy.

(97) Answer : (3)
Solution:
Sickle cell anaemia is a qualitative problem of synthesising an incorrectly functioning globin.

(98) Answer : (3)
Solution:
Sacred groves are found in Sarguja, Chanda and Bastar areas of Madhya Pradesh. Insects are the most species-rich taxonomic group among animals.
Core zone of biosphere reserve comprises an undisturbed and legally protected ecosystem.

(99) Answer : (3)
Solution:



(100) Answer : (1)
Solution:
In Honey bees, males are haploid and develop by means of parthenogenesis from an unfertilized egg. Males can have grandfather and grandsons.

(101) Answer : (2)
Solution:
Paul Ehrlich compared the rivets on the wings to key species that drive the major ecosystem functions.

(102) Answer : (3)
Solution:
Phenylketonuria is a classic example of pleiotropy.

(103) Answer : (2)
Solution:
Water hyacinth and water lily are pollinated by insects or wind. *Amorphophallus* is an insect pollinated plant.

(104) Answer : (2)
Solution:
α Thalassemia is controlled by two closely linked genes HBA 1 and HBA 2 on chromosome 16 of each parent and it is observed due to mutation or deletion of one or more of the four genes.

(105) Answer : (4)
Solution:
Colombia, located near the equator, has nearly 1400 species of birds while New York at 41°N has 105 species and Greenland at 71°N only 56 species and India has more than 1200 species of birds.

(106) Answer : (4)
Solution:
Birds show female heterogamety *i.e.* females produce two different types of gametes.

(107) Answer : (3)
Solution:
Types of gametes = 2^n (n = number of hybrid)
- AabbCcDD = 2^2 = 4
- aaBbCCDd = 2^2 = 4

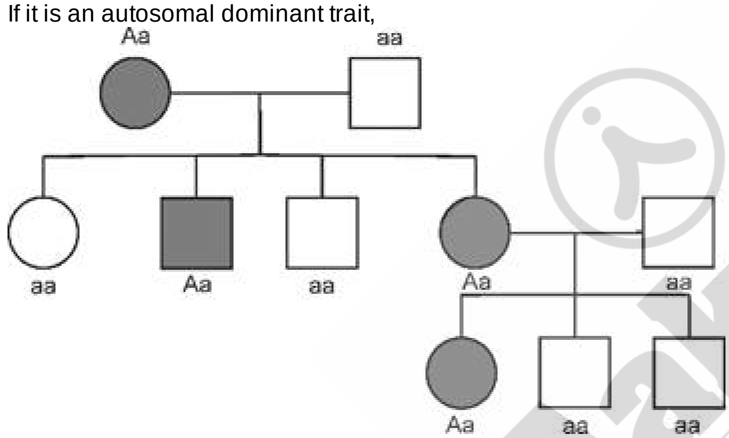
(108) Answer : (3)
Solution:
Dominance is not an autonomous feature of a gene or the product that it has information for.

(109) Answer : (2)
Solution:
In black pepper, remnant, persistent nucellus is called perisperm.

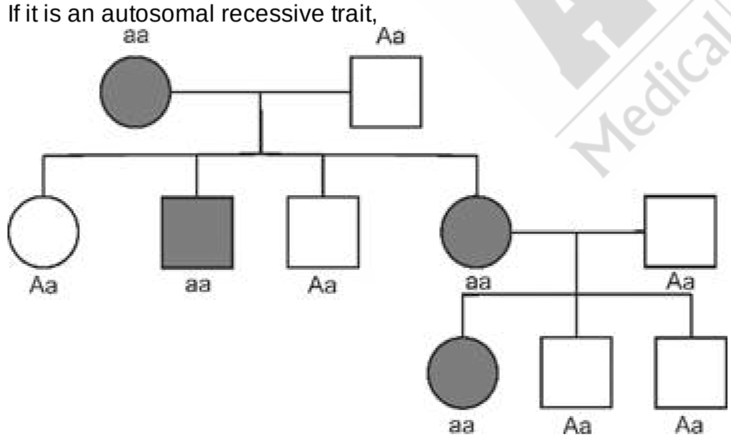
(110) Answer : (2)
Solution:
World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa. Biodiversity loss in a region may lead to increased variability in certain ecosystem processes such as pest and disease cycles.

(111) Answer : (3)
Solution:
Apomictic seeds do not show segregation of characters in hybrid progeny.

(112) Answer : (1)
Solution:
If it is an autosomal dominant trait,



or
If it is an autosomal recessive trait,

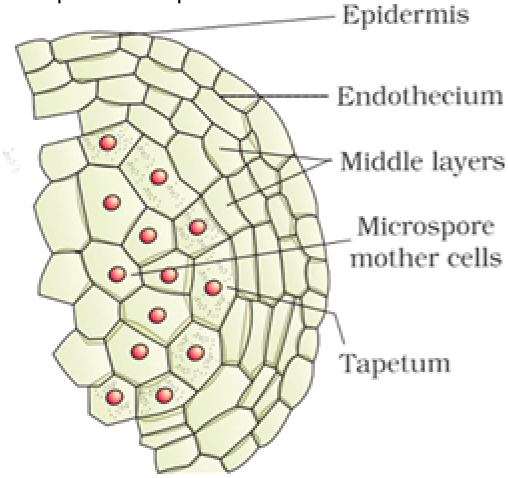


(113) Answer : (2)
Solution:
The number of contrasting characters studied by Mendel for his experiments was 7.

(114) Answer : (4)
Solution:
Haemophilia is an X-linked recessive disorder. In males affected with haemophilia, there will be only one X^h-chromosome which can be inherited by both daughter and grandson.

(115) Answer : (1)
Solution:
If the female parent produces unisexual flowers, there is no need for emasculation.

(116) Answer : (4)
Solution:
'd' represents tapetum which nourishes the developing pollen grains.



(117) Answer : (2)
Solution:
The number of ovules in an ovary may be one (wheat, paddy and mango) to many (papaya, water melon and orchids)

(118) Answer : (4)
Solution:

Parents: Genotype: ♂ I^AI^B × ♀ I^BI^B or I^Bi

	I^A	I^B
I^B	I^AI^B	I^BI^B
I^B	I^AI^B	I^BI^B

50% - AB
50% - B

or

	I^A	I^B
I^B	I^AI^B	I^BI^B
i	I^Ai	I^Bi

25% - AB
25% - A
50% - B

(119) Answer : (3)
Solution:
Filiform apparatus plays an important role in guiding the pollen tube into the synergid.

(120) Answer : (3)
Solution:
Endosperm persists in mature seeds of castor and coconut.

(121) Answer : (2)
Solution:
Mutagens – Artificially produce mutations like UV rays
Polyploidy – Often seen in plants

Aneuploidy – Arises due to non-disjunction of chromosomes
Frame-shift mutation – Deletions and insertions of base pairs of DNA

(122) Answer : (1)
Solution:
In *Ex situ* conservation, threatened animals and plants are taken out from their natural habitat and placed in special setting where they can be protected and given special care. Zoological parks, botanical gardens and wildlife safari parks serve this purpose.

(123) Answer : (2)
Solution:
In wind pollinated plants, pollen grains are light and non-sticky.
Colourful and fragrant flowers are found in insect pollinated plants.

(124) Answer : (4)
Solution:
Firewood, construction material and industrial products are narrowly utilitarian services. Aesthetic pleasure is a broadly utilitarian service.

(125) Answer : (3)
Solution:
Sacred groves are found in Aravalli hills of Rajasthan. Western ghats, Indo-Burma and Himalaya cover our country's exceptionally high biodiversity regions.

(126) Answer : (2)
Solution:
Tropical environments unlike temperate ones are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialization and lead to greater species diversity.

(127) Answer : (1)
Solution:
Tightly linked genes on the same chromosome show very few recombinations as compare to loosely linked ones.

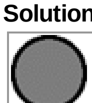
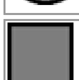
(128) Answer : (1)
Solution:
In tropical Amazonian rain forest, the biodiversity is as follows:

Taxa	**Number of species**
Birds	1300
Reptiles	378
Amphibians	427
Fishes	3,000

(129) Answer : (3)
Solution:
Self-incompatibility – Genetically controlled mechanism
Michelia – Apocarpous gynoecium
Generative cell – Spindle shaped with dense cytoplasm and a nucleus
Dioecious condition – Prevents autogamy as well as geitonogamy

(130) Answer : (4)
Solution:
Epicotyl – The cylindrical portion of embryonal axis above the level of cotyledons.
Phoenix dactylifera – Discovered during the archaeological excavation at king Herod's palace near dead sea.
Pollen viability – Pollen grains lose viability within 30 minutes of their release in rice and wheat.

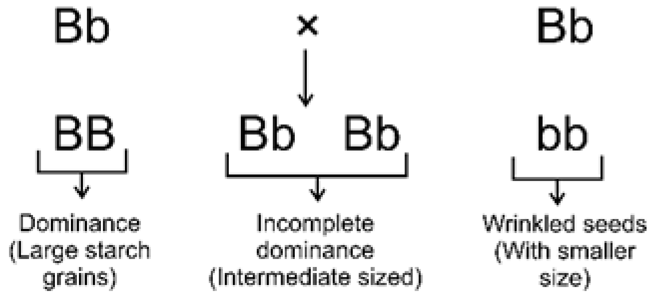
(131) Answer : (1)
Solution:

	–	Affected female
	–	Affected male
	–	Parents with affected male child with disease

(132) **Answer :** (3)

Solution:

Starch synthesis in pea seeds is an example of incomplete dominance, dominance and pleiotropy.



(133) **Answer :** (3)

Solution:

Many species extinctions in the last 500 years (Steller's sea cow, passenger pigeon) were due to overexploitation by humans.

(134) **Answer :** (1)

Solution:

ABO blood grouping shows multiple allelism.

(135) **Answer :** (2)

Solution:

Big and wrinkled tongue is observed in an individual inflicted with Down's syndrome.

ZOOLOGY

(136) **Answer :** (1)

Solution:

HIV contains 2 copies of single stranded RNA as its genetic material.

(137) **Answer :** (4)

Solution:

Marijuana, Hashish, Ganja, Charas – *Cannabis sativa*
Morphine – *Papaver somniferum*
Cocaine – *Erythroxylum coca*

(138) **Answer :** (2)

Solution:

Smoking increases carbon monoxide content in blood and reduces concentration of haembound oxygen as affinity of CO for Hb is much more than O_2.

(139) **Answer :** (4)

Solution:

Ascaris, an intestinal parasite causes ascariasis. Symptoms of this disease include internal bleeding, muscular pain, fever, anemia and blockage of intestinal passage.
Amoebic dysentery is characterised by excess mucus and blood clots in stools.
Ringworms is characterised by dry and scaly lesions on various parts of the body.

(140) **Answer :** (2)

Solution:

Diseases like malaria, filariasis, dengue, chikungunya *i.e.* diseases which spread through mosquito vector can be controlled by introducing fishes like *Gambusia* in ponds to feed upon mosquito larvae.
- ELISA is based upon antigen-antibody reaction.
- Vaccination is based upon memory of immune system.
- Indian population was approximately 350 million at the time of our independence and reached close to billion mark by 2000 and crossed 1.2 billion in May 2011.

(141) Answer : (1)
Solution:
Before Darwin, a French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs. Darwin said that true fitness is reproductive fitness. Mendel talked about inheritable factors influencing phenotype.

(142) Answer : (2)
Solution:
IUDs *i.e.* intrauterine devices are inserted by doctor or expert nurses in uterus through vagina. For example
- Non-medicated IUDs – Lippes loop
- Copper releasing IUDs – CuT, Cu7 and Multiload 375.
- Hormone releasing IUDs – Progestasert, LNG-20.

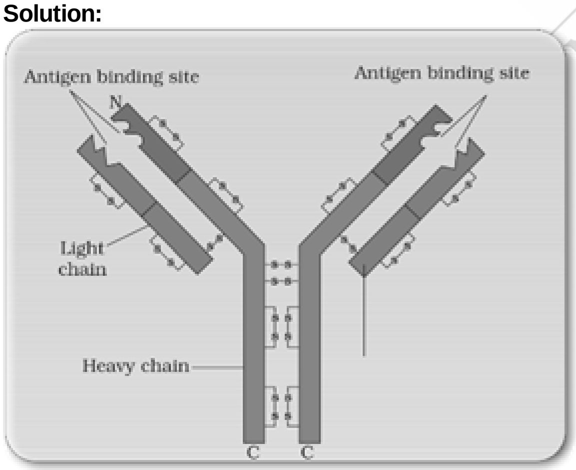
(143) Answer : (2)
Solution:
First mammals were like shrews. Their fossils were small sized. Mammals were viviparous and protected their unborn young inside the mother's body. They were more intelligent in sensing and avoiding danger at least.

(144) Answer : (1)
Solution:
Barrier methods of contraception are used along with spermicidal creams, jellies and foams to increase their contraceptive efficiency.
The possible ill-effects of contraceptive methods are nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding or even breast cancer, though not very significant, should not be totally ignored.

(145) Answer : (1)
Solution:
Recombinant DNA technology has allowed the production of antigenic polypeptides of pathogens in bacteria or yeast. Vaccines produced using this approach allow large scale production and hence great availability for immunisation. *E.g.* hepatitis-B vaccine produced from yeast.

(146) Answer : (2)
Solution:
The following diseases are vector borne *i.e.* require a vector for their transmission.
- Malaria
- Filariasis
- Dengue
- Chikungunya

(147) Answer : (2)
Solution:



In an antibody molecule
- Intra-heavy chains disulfide bonds = 8
- Inter-heavy chains disulfide bonds = 2
- Intra-light chain disulfide bonds = 4

• Number of disulfide bonds in between heavy and light chains = 2

(148) Answer : (3)
Solution:
Bobcat is a placental mammal.
Numbat (banded anteater) and bandicoot are Australian marsupials.

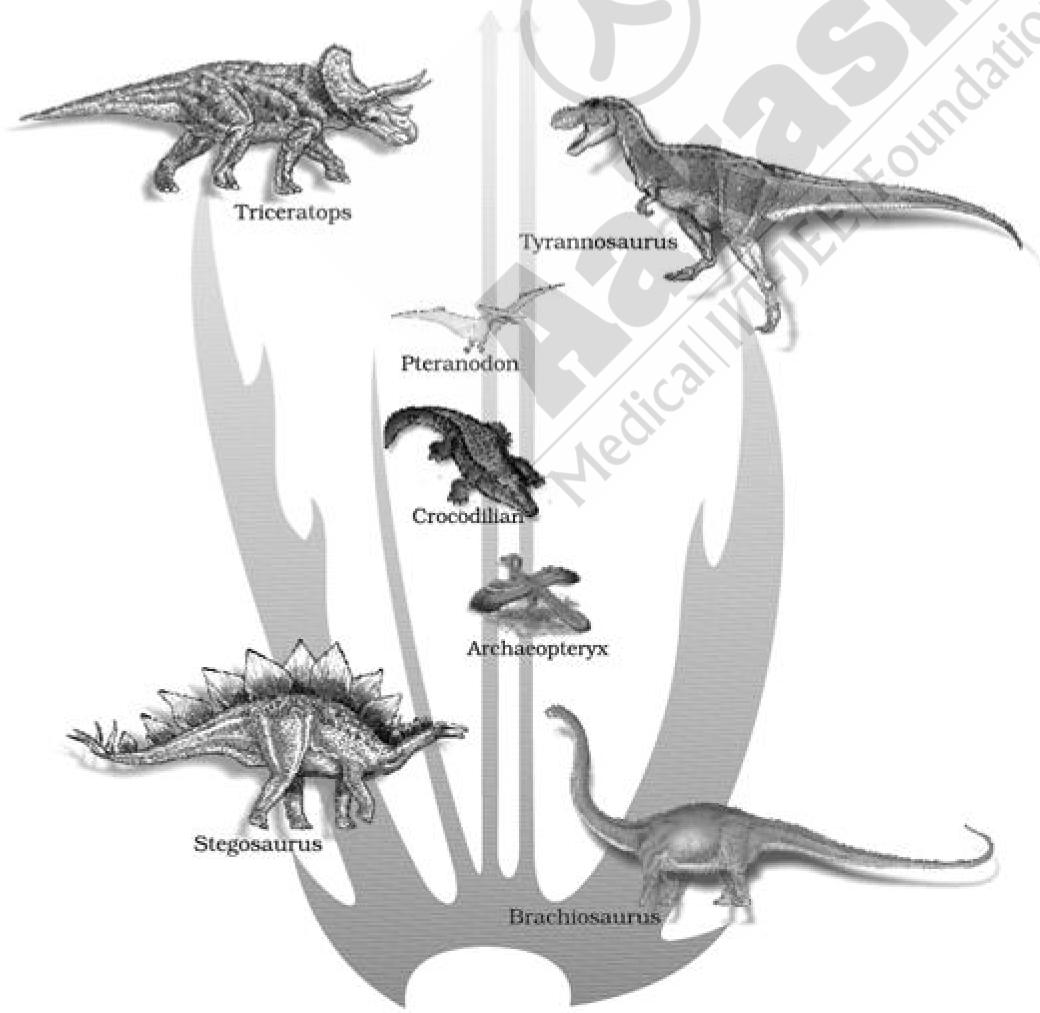
(149) Answer : (2)
Solution:
• According to World Health Organisation, reproductive health means a total well being in all aspects of reproduction *i.e.* physical, emotional, behavioural and social.
• Introduction of sex education in schools should be encouraged to provide right information to the youth.
• Indian laws permit legal adoption and it is as yet, one of the best methods for couples looking for parenthood.

(150) Answer : (4)
Solution:
Hallucinogens alter thoughts, feelings and perceptions.
Cocaine, obtained from *Erythroxylum coca* has a potent stimulating action on CNS. Its excessive dosage cause hallucinations. Other well-known plants with hallucinogenic properties are *Atropa belladona* and *Datura*.

(151) Answer : (2)
Solution:

Benign tumor	**Malignant tumor**
• Remains confined to the original location	• Invades and damages surrounding normal tissue
• Causes little damage	• Highly damaging
• Does not show metastasis	• Shows metastasis

(152) Answer : (1)
Solution:



(153) Answer : (3)
Solution:
The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. This type of immunity is called natural passive acquired immunity.

(154) Answer : (2)
Solution:
In vitro fertilisation is performed outside the body in almost similar conditions as that in the body.
GIFT is Gamete Intra-Fallopian Transfer
AI is Artificial Insemination
ZIFT is Zygote Intra Fallopian Transfer
IUT is Intra Uterine Transfer

(155) Answer : (1)
Solution:
Introduction of sex education in schools should be encouraged to provide right information to the young so as to discourage children from believing in myths and having misconceptions about sex related aspects. Also, proper information about reproductive organs, adolescence and related changes, safe and hygienic sexual practices, STDs, AIDs *etc* would help people lead a reproductively healthy life.

(156) Answer : (4)
Solution:
During post industrialisation period, the tree trunks became dark due to industrial smoke and soot. Under these conditions the white winged moths did not survive due to predators, dark winged or melanised moths survived.

(157) Answer : (1)
Solution:
MALT (Mucosa Associated Lymphoid Tissues) is a secondary lymphoid structure located within the lining of major tracts (respiratory, digestive and urinogenital tracts). It provides site for interaction of lymphocytes with antigens.

(158) Answer : (1)
Solution:
IUDs increase phagocytosis of sperms within uterus and the Cu ions released suppress sperm motility and the fertilising capacity of sperms.
The hormone releasing IUDs in addition make the uterus unsuitable for implantation and the cervix hostile to sperms.

(159) Answer : (1)
Solution:
In 1953, S.L Miller, an American scientist created similar conditions in a laboratory scale. He created electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at 800° C and observed formation of amino acids due to which chemical evolution was more or less accepted.

(160) Answer : (3)
Solution:
When our body encounters a pathogen for the first time it produces primary response. Subsequent encounter with same pathogen elicits highly intensified secondary or anamnestic response.
Both the responses are carried out with the help of two special types of lymphocytes present in blood *i.e.* B-lymphocytes and T- lymphocytes.

(161) Answer : (4)
Solution:
Administration of progestogen or progestogen- estrogen combination or IUDs within 72 hours of coitus have been found to be very effective as emergency contraceptives as they could be used to avoid possible pregnancy due to rape or casual unprotected intercourse.

(162) Answer : (3)
Solution:
Surgical interventions like vasectomy and tubectomy block gamete transport and thereby prevent conception.
If gamete transport is inhibited, no fertilisation or any further reproductive events occur.

(163) Answer : (3)
Solution:
Health increases longevity of people and reduces maternal and infant mortality.
When people are healthy, they are more efficient at work.

(164) Answer : (3)
Solution:
Embryological support for evolution was proposed by Ernst Heckel based upon the observation of certain features during embryonic stage common to all vertebrates that are absent in adult. This proposal was disapproved on careful study

performed by Karl Ernst von Baer. He noted that embryos never pass through the adult stages of other animals.

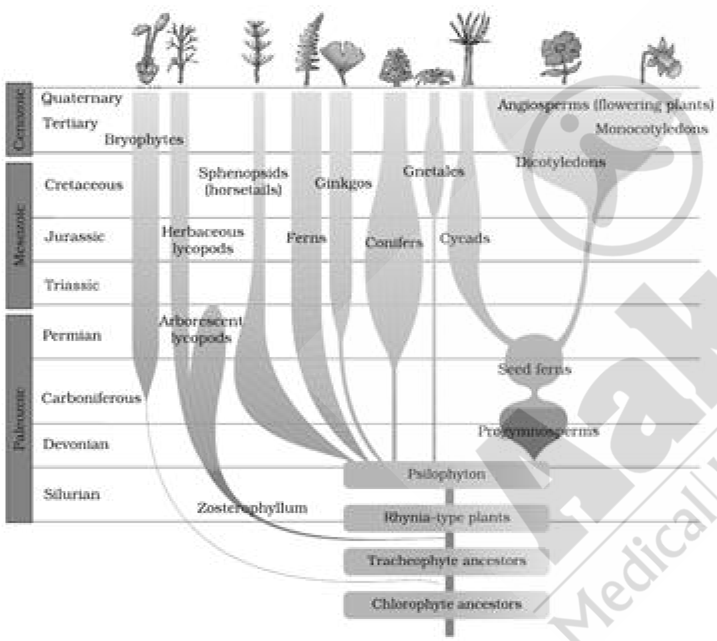
(165) Answer : (4)
Solution:
Forelimbs of whales and bats have similar anatomical structures, yet they perform different functions because the same structure developed along different directions due to adaptations to different needs.

(166) Answer : (4)
Solution:
Dryopithecus – More ape-like
Ramapithecus – Hairy and walked like gorillas and chimpanzees
Homo erectus – Probably ate meat
Homo habilis – Brain capacity of 650-800 cc

(167) Answer : (3)
Solution:
Streptococcus pneumoniae, *Haemophilus influenzae* – Pneumonia
Ascaris – Ascariasis
Entamoeba histolytica – Amoebic dysentery

(168) Answer : (3)
Solution:
The first non-cellular forms of life could have originated 3 billion years back.

(169) Answer : (4)
Solution:



(170) Answer : (1)
Solution:
If a woman has 28 days long menstrual cycle, the ovulation occurs on 14th day. Hence, couple should abstain from coitus from the day 10 to 17 of a 28 day menstrual cycle in order for woman to not get pregnant.

If menstrual cycle is of 30 days then ovulation will occur on 16th day, hence couple should abstain from coitus from the day 12 to 19 of the 30 days menstrual cycle.

(171) Answer : (2)
Solution:
The Big Bang theory attempts to explain to us the origin of Universe. It talks of a singular huge explosion unimaginable in physical terms. Other theories explain origin of life.

(172) Answer : (3)
Solution:
Physical barriers – Skin on our body along with mucus coating of GIT, respiratory and urinogenital tract.
Physiological barriers – Acid in stomach, saliva in mouth, tears from eyes.

(173) Answer : (2)

Solution:
Hugo deVries believed mutation caused speciation and hence called it saltation.
Convergent evolution is when different structures evolve for the same function and hence exhibit similarity.
Analogy is when structures are not anatomically similar though they perform similar functions.

(174) Answer : (4)
Solution:
Health for a very long time was considered as a state of body and mind where there was a balance of certain humors.
This is what early Greeks like Hippocrates as well as Indian Ayurveda system of medicine asserted.
Good humor hypothesis was disproved by William Harvey.

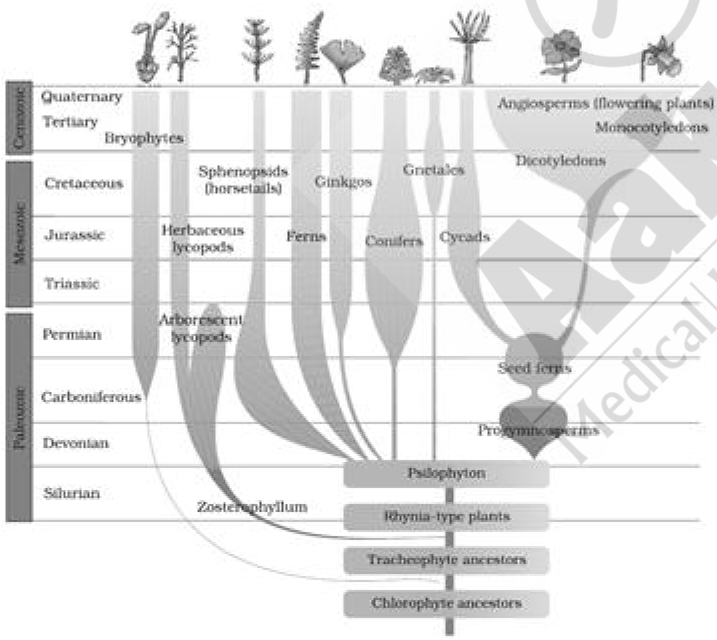
(175) Answer : (2)
Solution:
Theory of special creation states that the diversity was always the same since creation and will be the same in future also.

(176) Answer : (3)
Solution:
Different-aged rock sediments contains fossils of different life-forms who probably died during the formation of the particular sediment. Some of them appear similar to modern organisms.

(177) Answer : (1)
Solution:
Condoms, diaphragms, cervical caps and vaults are barrier methods of contraception. However, use of condoms has increased in recent years due to its additional benefit of protecting the user from contracting STDs and AIDS.
Hormone releasing IUDs make uterus unsuitable for implantation and the cervix hostile to sperms.

(178) Answer : (2)
Solution:
When we describe the story of this world we describe evolution as a process. On the other hand when we describe the story of life on Earth, we treat evolution as a consequence of processes called natural selection.

(179) Answer : (1)
Solution:



(180) Answer : (2)
Solution:
A monomeric antibody molecule is represented as H_2L_2 as there are 2 heavy chains and 2 light chains.